FrontEnd Theory Assignment - 03

ANSWERS –

1. What is the difference between frontend and backend JavaScript?

Frontend JavaScript run inside the browser and it is used for the things user see on the website, like button click, form check, changing text. Backend JavaScript run on the server with the help of Node.js. It handle things like database, login and API.

2. What is DOM?

DOM stands for Document Object Model. It is like a tree which browser makes from our HTML code. Every tag become a node in that tree. With JavaScript we can change the text, style, or add and delete the elements.
Example: document.querySelector("h1").style.color = "red";

3. What is event in JavaScript?

Event is any action done by user or browser on the page, for example click, double click, key press, page load. We write a function and attach it with the event so it run when that action happen.
Example: document.getElementById("btn").onclick = function() { alert("Clicked"); };

4. What is event loop in JavaScript?

JavaScript is single threaded so it do one task at one time. Event loop watch the call stack and the callback queue. When stack is empty, it take the waiting task from queue and send to stack. That is why setTimeout and API calls do not block the page.

5. What is hoisting in JavaScript?

Hoisting is the behaviour where variable and function declarations go to the top of the scope before code is executed. With var, the value is undefined if we use it before. Normal functions can be called before they are written. let and const give error if we use them before declaring.
Example: console.log(a); var a = 5; // undefined

6. What is event bubbling?

Event bubbling means when we click on inner element, the event first run on that element and then go up to the parent elements one by one. Like if button is inside a div, button event run first, after that div event run. To stop this we use stopPropagation().

7. What is the difference between named function, anonymous function and arrow function?

Named function is the function which have a name.
function greet() { console.log(“Hi”); }
Anonymous function does not have any name. We usually keep it in a variable.
const greet = function() { console.log(“Hi”); };
Arrow function is a short way of writing function using =>. It does not have its own “this” also.
const greet = () => console.log(“Hi”);

8. Explain for-of and for-in loop?

for-of is used to get the values from array or string.
for (let item of [“a”, “b”, “c”]) { console.log(item); } // a b c
for-in is used to get the keys of object, or index in case of array.
let user = {name: “Ravi”, age: 20};
for (let key in user) { console.log(key); } // name age

9. Explain any five array methods?

Five array methods:
push() - add new element in the end of array.
shift() - remove the first element of array.
map() - return new array after doing something on every element. [2,4,6].map(x => x + 1) gives [3,5,7].
filter() - return new array of only those elements which match the condition. [5,10,15].filter(x => x >= 10) gives [10,15].
includes() - check if the element is present in array or not, give true or false.

10. What is callback method?

Callback is a function which is given to another function as parameter, and it get called after the work is finished. Mostly used in async things like timer or fetching data.
Example: function hello(name, callback) { console.log(“Hello “ + name); callback(); } hello(“Ravi”, function() { console.log(“Done”); });